Cin-Yu Chen

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EDUCATION

M.Sc. in Mathematical Sciences *(GPA: 4.24 / 4.3)* 2022–2024
National Chengchi University, Taiwan
- Honorary Member, Phi Tau Phi Scholastic Honor Society
  (Awarded to up to 3% of master's graduates in each college.)

B.Sc. in Statistical Science *(GPA: 3.98 / 4.5)* 2017–2022
Department of Applied Mathematics, National Dong Hwa University, Taiwan
- Academic Excellence Award (Dean's List), Spring 2018 and Spring 2020

Minor in Arts & Design

COURSEWORK *(Graduate Level)*

Probability and Measure Theory
Stochastic Processes
Stochastic Calculus
Real Analysis

RESEARCH INTERESTS

Probability Theory, Stochastic Analysis, Martingale Theory, Mathematical Finance, Optimal Stopping, Hedging & Pricing.

RESEARCH EXPERIENCE

Master's Thesis: 2024
National Chengchi University

A Study of Discrete-Time Martingales with Applications to the Kalman Filter

Supervised by Prof. Shuenn-Jyi Sheu and Prof. Lung-Chi Chen

- Focusing primarily on discrete-time martingale theory
- Including a conceptual and exploratory study of the Kalman filter and its interpretation under partial information control problems.
- Full thesis available on my website.

Research Assistant: 2024–Jul. 2026

Mathematical Finance *National Central University*

Supervised by Prof. Shang-Yuan Shiu

- Develop and deliver expository presentations on mathematical finance from a probabilistic perspective, emphasizing motivation, intuition, and mathematical structure.
- Topics in discrete-time finite-market models include replication and martingale pricing of contingent claims; superhedging and optimal stopping for American claims; the fundamental theorems of asset pricing; and arbitrage-free price intervals in incomplete markets.
- Based on R.J. Williams, *Introduction to the Mathematics of Finance* and S.E. Shreve, *Stochastic Calculus for Finance*.

SEMINARS

Probability & Martingales 2022–2023

(M.Sc. Year 1) *National Chengchi University*

Supervised by Prof. Shuenn-Jyi Sheu

- Delivered proof-oriented seminar presentations on discrete-time martingales, stopping times, and measure-theoretic foundations of probability.
- Based on D. Williams, *Probability with Martingales* and R. Durrett, *Probability: Theory and Examples*.

Stochastic Calculus 2023–2024

(M.Sc. Year 2) *National Chengchi University*

Supervised by Prof. Shuenn-Jyi Sheu & Prof. Lung-Chi Chen

- Delivered proof-oriented seminar presentations on Brownian motion, continuous-time martingales, and stochastic integration.
- Based on J.-F. Le Gall, *Brownian Motion, Martingales, and Stochastic Calculus*.

SHORT COURSES

An Introduction to Diffusion Processes,
National Center for Theoretical Sciences (NCTS) , Taiwan Oct. 2025 (3 days)

A Brief Discussion on Brownian Motion and Related Processes with Applications,
National Center for Theoretical Sciences (NCTS) , Taiwan Jul. 2023 (2 weeks)

TEACHING EXPERIENCE

Teaching Assistant *National Chengchi University*

- Calculus I–II Fall 2023–Spring 2024

Teaching Assistant *National Dong Hwa University*

- Advanced Calculus I–II Fall 2021–Spring 2022
- Statistics Fall 2020
- Introduction to Probability Spring 2020

Selected Undergraduate Peer Tutor *National Dong Hwa University*
Fall 2020–Spring 2021

(Held weekly office hours to provide academic support to undergraduate students.)